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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49908 Roll No.

B. Tech. Examination, 2025-26

(Odd Semester)

COMPLEX ANALYSIS AND INTEGRAL TRANSFORMS

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : 8×1=8

   (a) Write C-R equations in polar form.

   (b) Define harmonic function.

   (c) State residue theorem.

   (d) Define singularity of the function with example.

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(e) Find the laplace transform of unit step function $u(t-a)$.

(f) Find the inverse laplace transform of $\frac{1}{2s-7}$.

(g) State the fourier integral theorem.

(h) Find the Z-transform of the sequence :

$\{8, 6, 3, -1, 0, 14, 5\}$

SECTION – B

2. Attempt any two parts of the following : 2×6=12

(a) Show that $e^x (x \cos y - y \sin y)$ is a harmonic function. By Milne-Thomson method, find the analytic function for which $e^x (x \cos y - y \sin y)$ is imaginary part.

(b) Evaluate the integral :

$$\int_0^{2\pi} \frac{1}{5-3\cos\theta}\, d\theta$$

(c) Using convolution theorem, find :

$$Z^{-1}\left[\frac{1}{s^3(s^2+1)}\right]$$

(d) Find the fourier sine transform of :

$$f(x) = \frac{e^{-ax}}{x}$$

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. 8×5=40

3. (a) Determine whether $\frac{1}{Z}$ is analytic or not?

(b) Evaluate :

$$\int_0^{1+i} (x^2 - iy)\, dz$$

along the path :

(i) $y = x$

(ii) $y = x^2$

(c) Use Cauchy's integral formula to evaluate :

$$\int_C \frac{z}{(z^2-3z+2)}\, dz$$

where C is the circle $|z-2| = \frac{1}{2}$.

4. (a) Expand $\frac{1}{z^2-3z+2}$ in the region :

(i) $|z|<1$

(ii) $1<|z|<2$

(b) Evaluate the following integral using residue theorem :

$$\int_C \frac{4-3z}{z(z-1)(z-2)}\,dz$$

where C is the circle $|z| = \frac{3}{2}$.

(c) Expand :

$$f(z) = \frac{1}{(z+1)(z+3)}$$

in Laurent series valid for :

(i) $1<|z|<3$

(ii) $|z|>3$

5. (a) Find the inverse laplace transform of :

$$\log\left(\frac{s+1}{s-1}\right)$$

(b) Find the laplace transform of :

$$t\,e^{-t}\sin 2t$$

(c) Using laplace transform, solve :

$$y'' + y = \sin 3t,$$
$$y(0) = 0,\ y'(0) = 0$$

6. (a) Express the function :

$$f(x) = \begin{cases} 1 & \text{when } |x| \le 1 \\ 0 & \text{when } |x| > 1 \end{cases}$$

as a fourier integral. Hence evaluate :

$$\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\,d\lambda$$

(b) Find the Z-transform of $\sin \alpha k$, $k \ge 0$.

(c) Find the inverse Z-transform of :

$$\frac{1}{(z-3)(z-2)},\quad 2<|z|<3$$

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